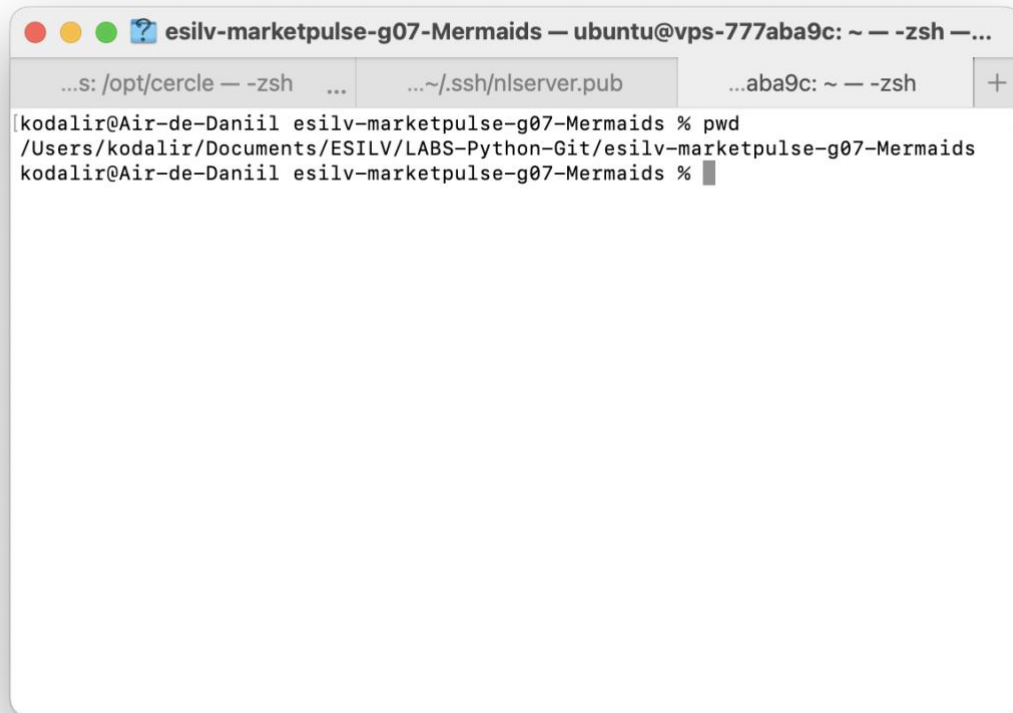


PART 5

5.1.



```
esilv-marketpulse-g07-Mermaids — ubuntu@vps-777aba9c: ~ — zsh — ...
...s: /opt/cercle — zsh  ...  ...~/.ssh/nlserver.pub  ...aba9c: ~ — zsh  +
[kodalir@Air-de-Daniil esilv-marketpulse-g07-Mermaids % pwd
/Users/kodalir/Documents/ESILV/LABS-Python-Git/esilv-marketpulse-g07-Mermaids
kodalir@Air-de-Daniil esilv-marketpulse-g07-Mermaids % █
```

5.2.

```
esilv-marketpulse-g07-Mermaids — ubuntu@vps-777aba9c: ~ — -zsh —...
...s: /opt/cercle — -zsh ... ...~/.ssh/nlserver.pub ...aba9c: ~ — -zsh +
[kodalir@Air-de-Daniil esilv-marketpulse-g07-Mermaids % pwd
/Users/kodalir/Documents/ESILV/LABS-Python-Git/esilv-marketpulse-g07-Mermaids
[kodalir@Air-de-Daniil esilv-marketpulse-g07-Mermaids % ls
CONTRIBUTING.md      data                requirements.txt
README.md             evidence            src
TEAM.md              labs
config               readiness
[kodalir@Air-de-Daniil esilv-marketpulse-g07-Mermaids % ls -la
total 56
drwxr-xr-x 14 kodalir  staff   448  7 oct 18:28 .
drwxr-xr-x  3 kodalir  staff    96  7 oct 17:49 ..
drwxr-xr-x 15 kodalir  staff   480  7 oct 18:30 .git
-rw-r--r--  1 kodalir  staff   178  7 oct 17:46 .gitignore
-rw-r--r--  1 kodalir  staff  2943  7 oct 17:46 CONTRIBUTING.md
-rw-r--r--  1 kodalir  staff 10270  7 oct 17:46 README.md
-rw-r--r--  1 kodalir  staff  1403  7 oct 18:28 TEAM.md
drwxr-xr-x  3 kodalir  staff    96  7 oct 17:46 config
drwxr-xr-x  3 kodalir  staff    96  7 oct 17:46 data
drwxr-xr-x  3 kodalir  staff    96  7 oct 17:46 evidence
drwxr-xr-x  6 kodalir  staff   192  7 oct 17:46 labs
drwxr-xr-x  3 kodalir  staff    96  7 oct 17:46 readiness
-rw-r--r--  1 kodalir  staff   133  7 oct 17:46 requirements.txt
drwxr-xr-x  3 kodalir  staff    96  7 oct 17:46 src
[kodalir@Air-de-Daniil esilv-marketpulse-g07-Mermaids % ]
```

5.3.

```
esilv-marketpulse-g07-Mermaids — ubuntu@vps-777aba9c: ~ — -zsh —...
...s: /opt/cercle — -zsh ... ...~/.ssh/nlserver.pub ...aba9c: ~ — -zsh +
[kodalir@Air-de-Daniil esilv-marketpulse-g07-Mermaids % cd data pwd ls
cd: too many arguments
[kodalir@Air-de-Daniil esilv-marketpulse-g07-Mermaids % cd data
[kodalir@Air-de-Daniil data % pwd
/Users/kodalir/Documents/ESILV/LABS-Python-Git/esilv-marketpulse-g07-Mermaids/data
[kodalir@Air-de-Daniil data % ls
sample
[kodalir@Air-de-Daniil data % cd sample
[kodalir@Air-de-Daniil sample % pwd
/Users/kodalir/Documents/ESILV/LABS-Python-Git/esilv-marketpulse-g07-Mermaids/data/sample
[kodalir@Air-de-Daniil sample % ls
bloomberg_reference_expected.json    instruments.json
bloomberg_reference_sample.json      prices.csv
[kodalir@Air-de-Daniil sample % cd ../../
[kodalir@Air-de-Daniil esilv-marketpulse-g07-Mermaids % pwd
/Users/kodalir/Documents/ESILV/LABS-Python-Git/esilv-marketpulse-g07-Mermaids
[kodalir@Air-de-Daniil esilv-marketpulse-g07-Mermaids % ]
```

PART 6

6.1.



```
esilv-marketpulse-g07-Mermaids — ubuntu@vps-777aba9c: ~ — -zsh —...
...s: /opt/cercle — -zsh  ...  ... ~/.ssh/nlserver.pub  ...aba9c: ~ — -zsh  +
[kodalir@Air-de-Daniil esilv-marketpulse-g07-Mermaids % cat data/sample/instrumen]
ts.json
{
  "instrument": {
    "ticker": "AAPL",
    "name": "Apple Inc.",
    "currency": "USD",
    "market": "NASDAQ"
  },
  "benchmark": {
    "ticker": "SP500",
    "name": "S&P 500",
    "currency": "USD",
    "market": "US"
  }
}
```

6.2.

```
esilv-marketpulse-g07-Mermaids — ubuntu@vps-777aba9c: ~ — -zsh —...
...s: /opt/cercle — -zsh ... ...~/.ssh/nlserver.pub ...aba9c: ~ — -zsh +
[kodalir@Air-de-Daniil esilv-marketpulse-g07-Mermaids % grep AAPL data/sample/pri]
ces.csv
2026-09-01,AAPL,249.20,251.50,248.00,250.00,38000000
2026-09-02,AAPL,250.80,252.80,249.60,251.30,39500000
2026-09-03,AAPL,249.60,251.30,248.40,249.80,41000000
2026-09-04,AAPL,251.30,253.60,250.10,252.10,42500000
2026-09-08,AAPL,251.90,253.90,250.70,252.40,44000000
2026-09-09,AAPL,254.00,255.70,252.80,254.20,38000000
2026-09-10,AAPL,254.30,256.60,253.10,255.10,39500000
2026-09-11,AAPL,254.20,256.20,253.00,254.70,41000000
2026-09-14,AAPL,256.60,258.30,255.40,256.80,42500000
2026-09-15,AAPL,257.20,259.50,256.00,258.00,44000000
2026-09-16,AAPL,256.70,258.70,255.50,257.20,38000000
2026-09-17,AAPL,259.20,260.90,258.00,259.40,39500000
2026-09-18,AAPL,259.30,261.60,258.10,260.10,41000000
2026-09-21,AAPL,259.10,261.10,257.90,259.60,42500000
2026-09-22,AAPL,261.10,262.80,259.90,261.30,44000000
2026-09-23,AAPL,261.20,263.50,260.00,262.00,38000000
2026-09-24,AAPL,263.00,265.00,261.80,263.50,39500000
2026-09-25,AAPL,262.60,264.30,261.40,262.80,41000000
2026-09-28,AAPL,263.30,265.60,262.10,264.10,42500000
2026-09-29,AAPL,264.50,266.50,263.30,265.00,44000000
2026-09-30,AAPL,266.00,267.70,264.80,266.20,38000000
kodalir@Air-de-Daniil esilv-marketpulse-g07-Mermaids %
```

PART 7

7.1.



A terminal window titled "esilv-marketpulse-g07-Mermaids — ubuntu@vps-777aba9c: ~ — zsh —...". The window has three tabs: "...s: /opt/cercle — zsh ...", "...~/.ssh/nlserver.pub", and "...aba9c: ~ — zsh". The terminal content shows a command prompt "kodalir@Air-de-Daniil esilv-marketpulse-g07-Mermaids %", followed by the command "python3 --version", the output "Python 3.14.5", and another command prompt "kodalir@Air-de-Daniil esilv-marketpulse-g07-Mermaids %" with a cursor.

```
esilv-marketpulse-g07-Mermaids % python3 --version
Python 3.14.5
esilv-marketpulse-g07-Mermaids %
```

7.2.

```
esilv-marketpulse-g07-Mermaids — ubuntu@vps-777aba9c: ~ — -zsh —...
...s: /opt/cercle — -zsh ... ...~/.ssh/nlserver.pub ...aba9c: ~ — -zsh +
[kodalir@Air-de-Daniil esilv-marketpulse-g07-Mermaids % git --version
git version 2.39.5 (Apple Git-154)
kodalir@Air-de-Daniil esilv-marketpulse-g07-Mermaids % █
```

7.3.

```
esilv-marketpulse-g07-Mermaids — ubuntu@vps-777aba9c: ~ — -zsh —...
...s: /opt/cercle — -zsh ... ...~/.ssh/nlserver.pub ...aba9c: ~ — -zsh +
[kodalir@Air-de-Daniil esilv-marketpulse-g07-Mermaids % python3 src/main.py
=== MarketPulse ===

Instrument
AAPL - Apple Inc.
Last price: 266.20 USD

Benchmark
SP500 - S&P 500
Last level: 6742.00

Period: 1 month
Interval: Daily

Observations
AAPL: 21
SP500: 21
kodalir@Air-de-Daniil esilv-marketpulse-g07-Mermaids % █
```

OPTIONAL



```
esilv-marketpulse-g07-Mermaids — ubuntu@vps-777aba9c: ~ — -zsh —...
...s: /opt/cercle — -zsh  ...  ... ~/.ssh/nlserver.pub  ...aba9c: ~ — -zsh  +
[kodalir@Air-de-Daniil esilv-marketpulse-g07-Mermaids % grep -c AAPL data/sample/
prices.csv
21
[kodalir@Air-de-Daniil esilv-marketpulse-g07-Mermaids % grep -c SP500 data/sample/
prices.csv
21
[kodalir@Air-de-Daniil esilv-marketpulse-g07-Mermaids % ]
```

```
esilv-marketpulse-g07-Mermaids — ubuntu@vps-777aba9c: ~ — -zsh —...
...s: /opt/cercle — -zsh ... ...~/.ssh/nlserver.pub ...aba9c: ~ — -zsh +
drwxr-xr-x 3 kodalir staff 96B 7 oct 17:46 data
drwxr-xr-x 3 kodalir staff 96B 7 oct 17:46 evidence
drwxr-xr-x 6 kodalir staff 192B 7 oct 17:46 labs
drwxr-xr-x 3 kodalir staff 96B 7 oct 17:46 readiness
-rw-r--r-- 1 kodalir staff 133B 7 oct 17:46 requirements.txt
drwxr-xr-x 3 kodalir staff 96B 7 oct 17:46 src
[kodalir@Air-de-Daniil esilv-marketpulse-g07-Mermaids % head -n 5 data/sample/prices.csv
date,ticker,open,high,low,close,volume
2026-09-01,AAPL,249.20,251.50,248.00,250.00,38000000
2026-09-01,SP500,6592.00,6615.00,6580.00,6600.00,0
2026-09-02,AAPL,250.80,252.80,249.60,251.30,39500000
2026-09-02,SP500,6607.00,6627.00,6595.00,6612.00,0
[kodalir@Air-de-Daniil esilv-marketpulse-g07-Mermaids % tails -n 5 data/sample/prices.csv
zsh: command not found: tails
[kodalir@Air-de-Daniil esilv-marketpulse-g07-Mermaids % tail -n 5 data/sample/prices.csv
2026-09-28,SP500,6720.00,6743.00,6708.00,6728.00,0
2026-09-29,AAPL,264.50,266.50,263.30,265.00,44000000
2026-09-29,SP500,6729.00,6749.00,6717.00,6734.00,0
2026-09-30,AAPL,266.00,267.70,264.80,266.20,38000000
2026-09-30,SP500,6740.00,6757.00,6728.00,6742.00,0
[kodalir@Air-de-Daniil esilv-marketpulse-g07-Mermaids %
```

```
esilv-marketpulse-g07-Mermaids — ubuntu@vps-777aba9c: ~ — -zsh —...
...s: /opt/cercle — -zsh ... ...~/.ssh/nlserver.pub ...aba9c: ~ — -zsh +
./.git/hooks/pre-receive.sample
./.git/hooks/prepare-commit-msg.sample
./.git/hooks/post-update.sample
./.git/hooks/pre-merge-commit.sample
./.git/hooks/pre-applypatch.sample
./.git/hooks/pre-push.sample
./.git/hooks/update.sample
./.git/hooks/push-to-checkout.sample
./.git/refs/heads/DBranche
./.git/refs/heads/main
./.git/refs/remotes/origin/DBranche
./.git/refs/remotes/origin/HEAD
./.git/refs/remotes/origin/main
./.git/index
./.git/packed-refs
./.git/COMMIT_EDITMSG
./.git/FETCH_HEAD
./TEAM.md
./data/sample/bloomberg_reference_sample.json
./data/sample/prices.csv
./data/sample/instruments.json
./data/sample/bloomberg_reference_expected.json
./src/main.py
[kodalir@Air-de-Daniil esilv-marketpulse-g07-Mermaids %
```

